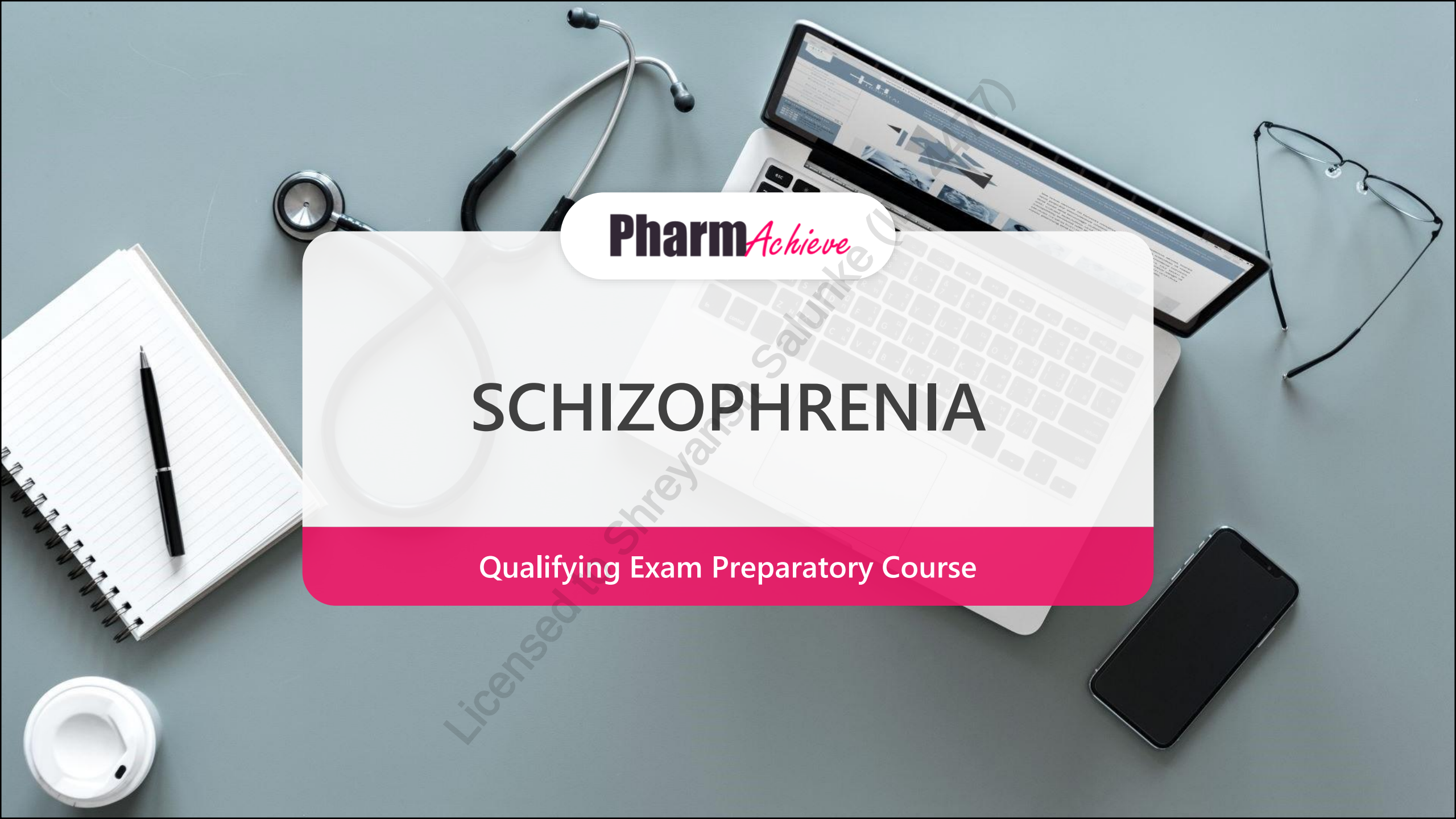




**Pharm***Achieve*

# SCHIZOPHRENIA

Qualifying Exam Preparatory Course

# COMPETENCIES COVERED IN THIS PRESENTATION...

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**Providing Care:  
Clinical Care**

**Providing Care:  
Distribution**

**Knowledge  
& Expertise**

**Communication  
& Collaboration**

**Leadership &  
Stewardship**

**Professionalism**

# LEGEND

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**ADR** Adverse Drug Reaction

**ANC** Absolute Neutrophil Count

**BID** Twice Daily

**BMI** Body Mass Index

**CBC** Complete Blood Count

**CNS** Central Nervous System

**D/C** Discontinue

**DSM** Diagnostic and Statistical Manual of Mental Disorders

**ECG** Electrocardiogram

**EPS** Extrapyramidal Symptoms

**FGA** First-Generation Antipsychotic

**IM** Intramuscular

**IV** Intravenous

**LSD** Lysergic Acid Diethylamide

**NMS** Neuroleptic Malignant Syndrome

**PCP** Phencyclidine

**PRN** As Needed

**QT INTERVAL** Time From Onset Of Q Wave To End Of T Wave

**SGA** Second-Generation Antipsychotic

**SL** Sublingual

**TD** Tardive Dyskinesia

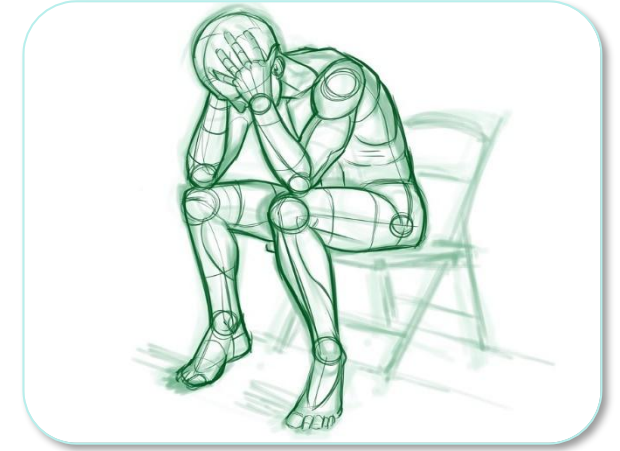
# DEFINITION

**Psychosis is a mental disorder that involves loss of contact with reality**

There are many mental illnesses that can include psychosis as a symptom

**Schizophrenia** refers to a *heterogeneous syndrome* with psychosis as a hallmark feature

- It causes a significant decrease in an individual's ability to function
- Symptoms and length of illness may vary from person to person



# PATHOPHYSIOLOGY

## MULTIFACTORIAL AND NOT CLEARLY DEFINED

4 Key Dopaminergic Tracts: Tracts 2 and 3 are implicated in schizophrenia

1. Nigrostriatal – Dopamine receptor blockade → EPS
2. Mesolimbic – ↑ dopamine → positive symptoms
3. Mesocortical – ↓ dopamine → negative symptoms
4. Tuberoinfundibular – Dopamine blockade → increased prolactin

### Positive symptoms

- Excess or distortion of normal function
- Associated with dopamine receptor *hyperactivity* (mesolimbic)

### Negative symptoms

- Decrease or absence of normal function
- Related to dopamine receptor *hypoactivity* (mesocortical)

# CLINICAL PRESENTATION

| Schizophrenic Symptoms                                                                                                                      |                                                                                                                                                                                                                                           |                                                                                                                                              |                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Positive                                                                                                                                    | Negative                                                                                                                                                                                                                                  | Cognitive                                                                                                                                    | Mood                                                                                                                                          |
| <ul style="list-style-type: none"><li>• Delusions</li><li>• Hallucinations</li><li>• Disorganization (speech, behaviour, thought)</li></ul> | <ul style="list-style-type: none"><li>• Affective flattening and/or apathy</li><li>• Alogia (aphasia)</li><li>• Anhedonia (lack of pleasure)</li><li>• Avolition (lack of motivation)</li><li>• Asociality or social withdrawal</li></ul> | <ul style="list-style-type: none"><li>• Impaired attention</li><li>• Impaired working memory</li><li>• Impaired executive function</li></ul> | <ul style="list-style-type: none"><li>• Depression</li><li>• Anxiety</li><li>• Aggression</li><li>• Hostility</li><li>• Suicidality</li></ul> |

| Residual Features (varied symptoms and severity)                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"><li>1. Anxiety, suspiciousness, <b>lack of volition/insight/judgment</b></li><li>2. Difficulty <b>living independently</b> in community</li><li>3. Difficulty <b>learning from mistakes</b> in social situations</li><li>4. Difficulty understanding importance of treatment (<b>Non-compliance: ↑ risk of relapse</b>)</li></ol> |

# RISK FACTORS

|                                                                                    |                                                                                       |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|   | Low birth weight                                                                      |
|   | Genetics<br>(risk of schizophrenia is 10% if first-degree relative has schizophrenia) |
|   | Advanced paternal age                                                                 |
|  | Substance abuse                                                                       |

## Drug Causes

- Many street drugs can lead to/worsen psychosis symptoms
  - Cocaine
  - Lysergic acid diethylamide (LSD)
  - Methamphetamine
  - Marijuana
  - Ecstasy
  - Phencyclidine (PCP)
  - Ketamine
  - Anticholinergics
  - Dopamine agonists
  - Levodopa

Maternal risk factors: obesity, viral illness, malnutrition, exposure to cats infected with toxoplasmosis among others

# DIAGNOSIS

---

## DSM-5 CRITERIA

1. Mental status
2. Physical and neurological status
3. Complete family and social history
4. Laboratory workup (medical/substance causes)

- A. 2 or more of the following for at least 1 month:
  - Delusions, hallucinations, disorganized speech, grossly disorganized or catatonic behaviour, negative symptoms
  - **At least 1** must be delusions, hallucinations or disorganized speech
- B. Social and/or occupational dysfunction (work, interpersonal relations, or self-care)
- C. Duration: signs persist for **at least 6 months**, with at least 1 month of symptoms that meet criteria A
- D. Schizoaffective or mood disorder has been excluded
- E. Disorder is not due to a medical disorder or substance
- F. If a history of autism spectrum disorder or communication disorder of childhood onset is present, there must be symptoms of hallucinations or delusions present for **at least 1 month**

# GOALS OF THERAPY

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- ✓ Eliminate/reduce psychotic signs and symptoms
- ✓ Reduce frequency of future episodes
- ✓ Prevent relapse
- ✓ Minimize adverse reactions related to therapy
- ✓ Manage risks of metabolic syndrome, substance abuse, depression and other comorbidities
- ✓ Promote adherence by implementing strategies to improve patient's ability to take medications
- ✓ Reduce risk of harm and social isolation

# NON-PHARMACOLOGICAL MEASURES

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Psychosocial rehabilitation  
which may include:

Basic living skills

Social skills training

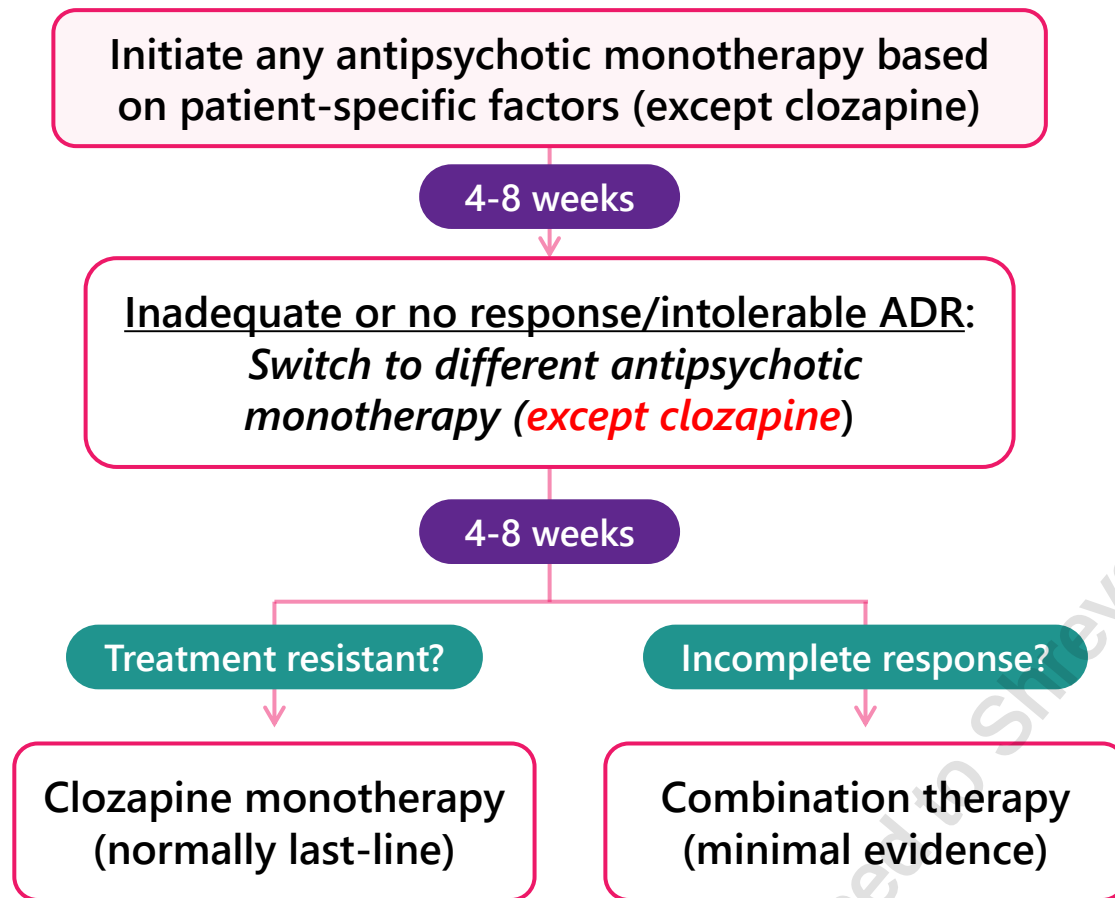
Basic education

Work programs

Supported housing

Involving family in the care and life of the patient can ↓ *rehospitalization* and *improve functioning* in the community

# TREATMENT ALGORITHM



1. **Titrate therapy** over few days to weeks until desired dose (start low, go slow)
2. Assess for comorbidities (e.g. depression, substance abuse). If there is poor response, consider assessment of medication adherence
3. Continue therapy for...
  - At least **1-2 years** post-remission of first episode
  - At least **2-5 years** post remission of first episode for complicated case
  - At least **5 years** for more than 2 episodes
  - Possibly an **indefinite period**

## Pair with non-pharmacological measures:

- Psychoeducation, simplified dosing regimens (once daily vs BID - IR vs LA formulations), blister packs, dosettes, care giver support

# PHARMACOLOGICAL MEASURES

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Acute and maintenance treatment: **first-and second-generation antipsychotics**

- All have similar efficacy, agent should be chosen **based on patient specific factors** (e.g. tolerability, side effects)
- First-line: **second-generation antipsychotics** (better tolerability)
- **First-generation antipsychotics** (typical or conventional)
  - **High-potency:** haloperidol, flupenthixol, fluphenazine, pimozide
  - **Mid-potency:** loxapine, perphenazine, zuclopenthixol
  - **Low-potency:** chlorpromazine, methotrimeprazine
- **Second-generation antipsychotics** (atypical)
  - Aripiprazole, asenapine, brexpiprazole, cariprazine, clozapine, lurasidone, olanzapine, paliperidone, quetiapine, risperidone, ziprasidone
  - Greater 5HT<sub>2A</sub> affinity relative to D<sub>2</sub>

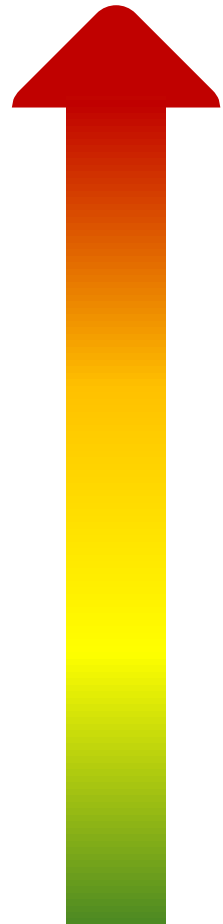
**\*\* All antipsychotics have been associated with increased risk of mortality and cerebrovascular events (e.g. stroke) in patients with dementia \*\***

# PHARMACOLOGICAL MEASURES

| RECEPTOR ANTAGONIZED            | CLINICAL EFFECTS                                                                                                                                                                                                                                                                                                                                                                                                                           | ANTIPSYCHOTICS MOST RELEVANT                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dopamine (D <sub>2</sub> )      | <ul style="list-style-type: none"> <li>Antagonism of D<sub>2</sub> receptors is the basic mechanism of action of all antipsychotics since increased D<sub>2</sub> activity in the mesolimbic tract of the brain may be related to positive symptoms of schizophrenia</li> <li>Excessive dopamine blockade in other dopaminergic tracts of the brain may be associated with EPS, tardive dyskinesia (TD), and hyperprolactinemia</li> </ul> | <ul style="list-style-type: none"> <li>All antipsychotics have D<sub>2</sub> antagonism activity (except aripiprazole, cariprazine and brexpiprazole, which are partial dopamine agonists and less potent than endogenous dopamine and result in an overall antagonistic effect)</li> <li>FGAs have higher affinity than SGAs, especially <b>high potency</b> FGAs</li> </ul> |
| Histaminic (H <sub>1</sub> )    | <ul style="list-style-type: none"> <li>Sedation, appetite stimulation, weight gain</li> </ul>                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>Common with <b>low potency</b> FGAs, clozapine, quetiapine, olanzapine</li> </ul>                                                                                                                                                                                                                                                      |
| Muscarinic (M <sub>1</sub> )    | <ul style="list-style-type: none"> <li>Anticholinergic effects (e.g. dry mouth, constipation)</li> </ul>                                                                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>Common with <b>low potency</b> FGAs - <b>avoid</b> in those with significant cognitive impairment, dementia or delirium and in the elderly</li> </ul>                                                                                                                                                                                  |
| Noradrenergic (Alpha-1)         | <ul style="list-style-type: none"> <li>Orthostatic hypotension</li> </ul>                                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>Common with <b>low-potency</b> FGAs, clozapine, olanzapine, quetiapine, risperidone, ziprasidone</li> </ul>                                                                                                                                                                                                                            |
| Serotonin (5-HT <sub>2A</sub> ) | <ul style="list-style-type: none"> <li>Antagonism of 5-HT<sub>2A</sub> receptors may increase dopaminergic activity in the mesocortical pathway (improving negative symptoms of schizophrenia) and the nigrostriatal pathway (decreasing risk of EPS)</li> </ul>                                                                                                                                                                           | <ul style="list-style-type: none"> <li>SGAs have significant 5-HT<sub>2A</sub> antagonist activity</li> </ul>                                                                                                                                                                                                                                                                 |

# PHARMACOLOGICAL MEASURES

## FIRST-GENERATION (TYPICAL) ANTIPSYCHOTICS



High-Potency: higher rate of EPS, NMS, and elevated prolactin levels

- Haloperidol
- Flupenthixol
- Fluphenazine
- Pimozide

Mid-Potency:

- Loxapine
- Perphenazine
- Zuclopenthixol

Low-Potency: higher rates of sedation, cardiovascular effects, anticholinergic effects, and weight gain

- Chlorpromazine
- Methotrimeprazine

All FGAs are **equally efficacious** when used in equipotent dose

Potency is determined by dopamine D<sub>2</sub>-receptor binding affinity

Consider using **mid-potency agents**

Selection should focus on ↓ **side effects and concurrent medical/psychiatric disorders**

# PHARMACOLOGICAL MEASURES

## FIRST-GENERATION (TYPICAL) ANTIPSYCHOTICS

| Name           | Special features                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Haloperidol    | <ul style="list-style-type: none"><li>• Available as injectable and PO formulations</li><li>• IM formulation has been used for psychotic agitation</li><li>• Lactate: Immediate acting injection Q4-8H PRN</li><li>• Decanoate: Long-acting injection Q4 weeks</li><li>• Can use in combination with promethazine to manage psychosis-induced aggression</li></ul>                                                                                                                                                                                                                            |
| Zuclopenthixol | <ul style="list-style-type: none"><li>• Available as injectable and PO formulations</li><li>• Acetate: Administered IM Q2-3 days<ul style="list-style-type: none"><li>• Peak serum level in 24-48 hours, declining to one-third of peak concentration in 72 hours</li><li>• This can reduce number of injections required in severe acute agitation and/or aggression</li><li>• Do not use in antipsychotic-naïve patients with first episode psychosis</li><li>• Intended for use for acute psychosis for up to 2 weeks</li></ul></li><li>• Decanoate: LA injection, IM Q2-4 weeks</li></ul> |

***Switch to the oral route as soon as practical, due to greater risk of EPS with IM administration***

# PHARMACOLOGICAL MEASURES

## SECOND-GENERATION (ATYPICAL) ANTIPSYCHOTICS

| Name                     | Special features                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ziprasidone<br>(Zeldox®) | <ul style="list-style-type: none"><li>• Moderate inhibition of synaptic reuptake of serotonin and norepinephrine</li><li>• BID dosing</li><li>• Must be <b>given with food</b> (minimum <b>500 calories</b>) to maximize absorption → issue in patients with dyslipidemia, metabolic syndrome</li><li>• <b>Weight gain neutral</b></li><li>• Baseline ECG done before initiating therapy</li><li>• <b>Avoid</b> with other QTc prolonging medications or history of QTc prolongation, recent MI, uncompensated heart failure → <b>D/C if QT &gt; 500 msec</b></li><li>• Ensure hypomagnesemia, hypokalemia are corrected</li></ul> |
| Lurasidone<br>(Latuda®)  | <ul style="list-style-type: none"><li>• Must be <b>given with food to maximize absorption</b> (minimum <b>350 calories</b>, regardless of fat)</li><li>• <u>Side effects</u>: Nausea common at initiation and with dose increases and can be persistent; it is often worst 2–3 h post dose (time of peak plasma levels) and not related to presence or absence of food.</li><li>• <b>Weight gain neutral</b></li></ul>                                                                                                                                                                                                             |
| Asenapine<br>(Saphris®)  | <ul style="list-style-type: none"><li>• BID dosing</li><li>• Sublingual tablet: Instruct patients to handle the tablet with dry hands, to place it under the tongue immediately, to allow it to dissolve completely (do not chew or swallow)</li><li>• <b>Should not eat or drink for minimum 10 minutes after administration</b> (food/drink may reduce bioavailability)</li><li>• May have an unpleasant taste and cause tongue numbness/tingling, usually resolves in 1 hour</li></ul>                                                                                                                                          |

# PHARMACOLOGICAL MEASURES

## SECOND GENERATION (ATYPICAL) ANTIPSYCHOTICS

| Name                        | Special features                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Olanzapine<br>(Zyprexa®)    | <ul style="list-style-type: none"><li>• Very sedating and negative metabolic effects, <b>Large weight gain associated</b></li><li>• Available in short acting IM injection and oral dissolvable tablet</li><li>• Primarily metabolized by CYP1A2 and CYP2D6, <b>Smoking cigarettes induces CYP1A2 and can lower levels of olanzapine</b></li><li>• <b>DO NOT USE</b> with parenteral formulations of benzodiazepines due to reports of cardiac and respiratory problems</li></ul> |
| Quetiapine<br>(Seroquel®)   | <ul style="list-style-type: none"><li>• Available as IR (immediate release) and XR formulations (extended release)</li><li>• Generally dosed at bedtime</li><li>• May be abused for sedative and anxiolytic properties due to antihistaminic and anticholinergic properties</li></ul>                                                                                                                                                                                             |
| Risperidone<br>(Risperdal®) | <ul style="list-style-type: none"><li>• High frequency of EPS → benztropine may be used to reduce EPS</li><li>• <b>Available as long-acting IM injectable (Risperdal Consta®)</b></li></ul>                                                                                                                                                                                                                                                                                       |
| Paliperidone<br>(Invega®)   | <ul style="list-style-type: none"><li>• Active metabolite of risperidone</li><li>• Less orthostatic hypotension and weight gain and more insomnia compared to risperidone</li><li>• Not metabolized by liver, thus minimal drug interactions</li><li>• <b>Renal dose adjustment required</b></li><li>• <b>Available as long-acting IM injectable (Invega Sustenna® and Invega Trinza®)</b></li></ul>                                                                              |

# PHARMACOLOGICAL MEASURES

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## SECOND-GENERATION (ATYPICAL) ANTIPSYCHOTICS

**Clozapine:** Only antipsychotic to demonstrate **therapeutic superiority**, however, it is **reserved for treatment-resistant schizophrenia** due to **severe side effects**

- **Treatment-resistant schizophrenia:** failure to respond ( $\leq 20\%$  reduction in positive symptoms) to 2 or more adequate trials ( $\geq 6$  weeks) of non-clozapine antipsychotic medication
- Side Effects profile:
  - Agranulocytosis -patients must undergo regular blood tests (CBC) to monitor this
  - Myocarditis
  - Orthostatic hypotension
  - Seizures
  - Hypersalivation
  - Heart failure
  - Constipation-which can worsen to a fatal intestinal obstruction
- Must be **titrated** to mitigate risk of orthostatic hypotension, sedation, and seizures
- Metabolized by CYP1A2 (caution with concomitant cigarette smoking) and CYP3A4

# PHARMACOLOGICAL MEASURES

## Clozapine: Monitoring

### SECOND-GENERATION (ATYPICAL) ANTIPSYCHOTICS

| GREEN<br>(Normal values for clozapine treatment)                                                                                                                                                                                |                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ANC $\geq 2.0 \times 10^9$ /L                                                                                                                                                                                                   | <ul style="list-style-type: none"><li>Continue to dispense CLOZARIL®</li><li>Monitor as follows for eligible patients: – Weekly for the first 26 weeks – Every 2 weeks for the next 26 weeks – Every 4 weeks as of 52 weeks</li></ul>                                                     |
| YELLOW alert<br>Particular attention should be paid if patient presents with the following: Any flu-like complaints or other symptoms that might suggest infection (i.e., fever, sore throat, or any other signs of infection). |                                                                                                                                                                                                                                                                                           |
| ANC is $\geq 1.5 \times 10^9$ /L < $2.0 \times 10^9$ /L                                                                                                                                                                         | <ul style="list-style-type: none"><li>Hematological monitoring at least twice a week until ANC stabilize or increase</li><li>Continue to dispense CLOZARIL®</li></ul>                                                                                                                     |
| RED ALERT                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                           |
| ANC < $1.5 \times 10^9$ /L                                                                                                                                                                                                      | <ul style="list-style-type: none"><li>Immediately withhold CLOZARIL® and monitor patient closely</li><li>If the drug is discontinued due to neutropenia, monitoring should be conducted at least twice a week until ANC is normal (ANC is <math>\geq 2.0 \times 10^9</math> /L)</li></ul> |

# PHARMACOLOGICAL MEASURES

## SECOND/THIRD GENERATION (ATYPICAL) ANTIPSYCHOTICS

**Mechanism of Action:** partial agonist activity at D<sub>2</sub> and 5-HT<sub>1A</sub> receptors and potent antagonism activity at 5-HT<sub>2A</sub> receptors (potential efficacy in treating negative and depressive symptoms)

| Name                         | Special features                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aripiprazole<br>(Abilify®)   | <ul style="list-style-type: none"><li>• Best taken in the morning → can cause activating side effects (e.g. insomnia)</li><li>• Maximum 30 mg/day</li><li>• No greater efficacy seen with higher doses</li><li>• Available as long-acting IM injectable (Abilify Maintena® – monthly)<ul style="list-style-type: none"><li>• Ensure patient tolerates oral formulation before initiating IM</li></ul></li></ul>                                                      |
| Brexipiprazole<br>(Rexulti®) | <ul style="list-style-type: none"><li>• Relatively lower risk of EPS and metabolic disorders (e.g. weight gain)</li><li>• Not affected by smoking</li><li>• Also indicated as adjunctive therapy in major depressive disorder</li></ul>                                                                                                                                                                                                                              |
| Cariprazine<br>(Vraylar®)    | <ul style="list-style-type: none"><li>• Newest antipsychotic on the Canadian market</li><li>• Has a long half-life → may take weeks for side effects to manifest after starting therapy</li><li>• Also indicated for the treatment of Bipolar I disorder</li><li>• Extensive metabolism by CYP3A4 → contraindicated in combo with moderate-strong CYP3A4 inducers/inhibitors</li><li>• Akathisia, hypertension and EPS were most common in clinical trials</li></ul> |

# PHARMACOLOGICAL MEASURES - DRUG INTERACTIONS

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## SECOND GENERATION (ATYPICAL) ANTIPSYCHOTICS

- Most antipsychotics are metabolized by CYP3A4
  - **Inhibitors** of CYP3A4 (e.g. clarithromycin, erythromycin, grapefruit juice, ketoconazole) can increase levels of the antipsychotics
  - **Inducers** of CYP3A4 (e.g. carbamazepine, phenytoin, rifampin) can decrease levels of antipsychotics
- Olanzapine is primarily metabolized by CYP1A2 and 2D6
  - **Inhibitors** of CYP1A2 or CYP2D6 (e.g. fluvoxamine or paroxetine) may increase olanzapine levels
  - **Inducers** of CYP1A2 (e.g. polycyclic aromatic hydrocarbons in cigarette smoking) may decrease olanzapine levels
    - This is also evident with clozapine

# SIDE EFFECT PROFILES

|              | Weight gain | ↑ Blood sugar | Dyslipidemia | Hyperprolactinemia | EPS*     | Sedation* |
|--------------|-------------|---------------|--------------|--------------------|----------|-----------|
| Clozapine    | HIGH        | HIGH          | HIGH         | RARE               | RARE     | HIGH      |
| Olanzapine   | HIGH        | HIGH          | HIGH         | LOW                | RARE-LOW | HIGH      |
| Quetiapine   | MED         | HIGH          | MED          | RARE               | RARE     | HIGH      |
| Risperidone  | MED         | MED           | MED          | HIGH               | MED      | MED       |
| Paliperidone | LOW-MED     | LOW-MED       | LOW-MED      | HIGH               | MED      | MED       |
| Aripiprazole | LOW         | LOW           | LOW          | RARE               | LOW      | LOW       |
| Ziprasidone  | LOW         | LOW           | LOW          | LOW                | LOW      | LOW-MED   |
| Lurasidone   | LOW         | LOW           | LOW          | LOW                | LOW      | LOW-MED   |
| Asenapine    | LOW-MED     | LOW           | LOW          | -                  | -        | LOW-MED   |

\*Sedation: Take agent at bedtime

\*EPS: Use/switch to SGA. Give benztropine or diphenhydramine to treat symptoms

# SIDE EFFECT PROFILES

|              | QT Prolonging | Seizures | Anticholinergic | Postural hypotension |
|--------------|---------------|----------|-----------------|----------------------|
| Clozapine    | LOW-MED       | HIGH     | HIGH            | HIGH                 |
| Olanzapine   | LOW-MED       | LOW      | MED-HIGH        | HIGH                 |
| Quetiapine   | LOW-MED       | LOW      | HIGH            | MED                  |
| Risperidone  | LOW-MED       | LOW      | LOW-MED         | MED                  |
| Paliperidone | MED           | LOW      | LOW-MED         | LOW                  |
| Aripiprazole | LOW-MED       | LOW      | RARE            | LOW                  |
| Ziprasidone  | HIGH          | -        | MED             | MED                  |
| Lurasidone   | -             | LOW      | LOW             | LOW-MED              |
| Asenapine    | LOW-MED       | LOW      | LOW             | MED                  |

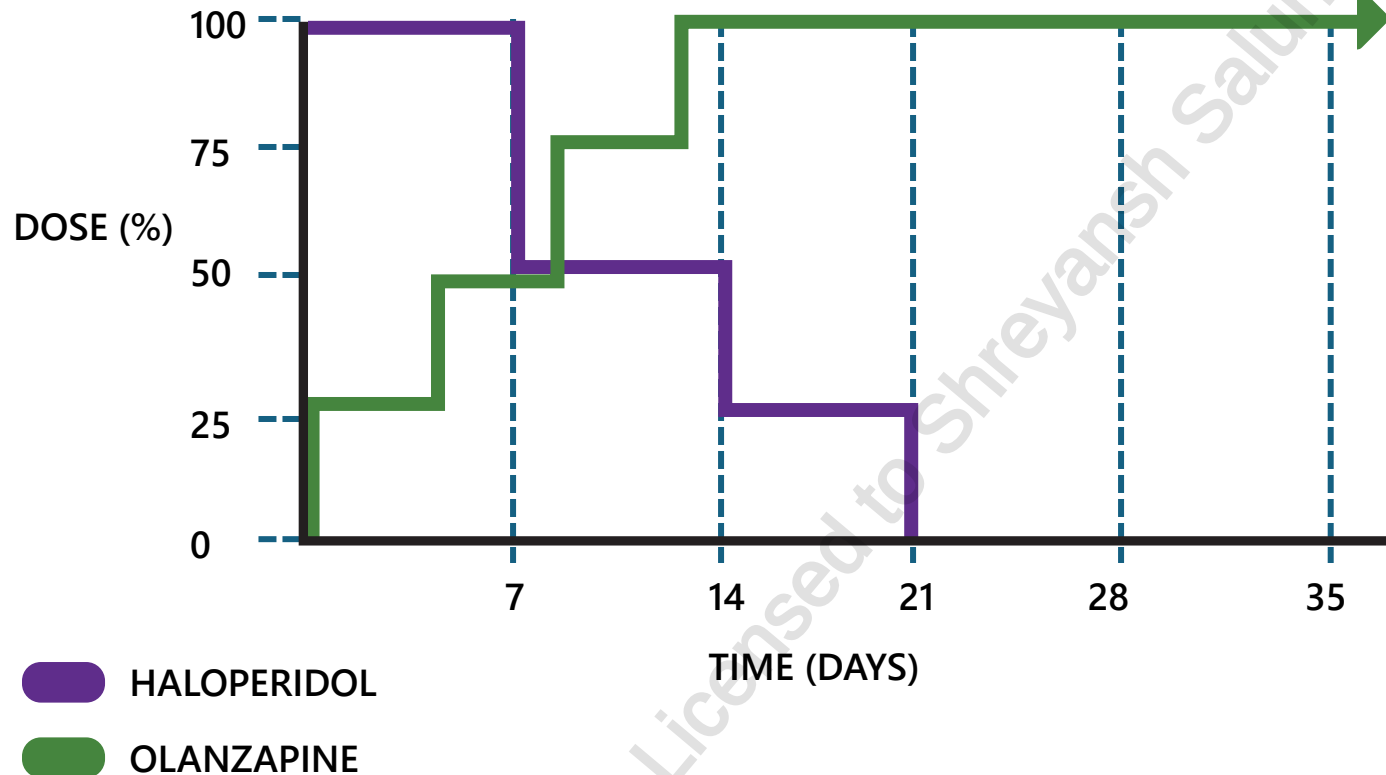
# NEUROLEPTIC MALIGNANT SYNDROME (NMS)

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- Life-threatening side effect that may occur with any antipsychotic but more commonly occurs with FGAs
- May occur at any time during treatment
- **Symptoms:** Fever, mental status changes, increased creatine kinase, rigidity, autonomic dysfunction, increased WBC counts, and changes in consciousness
- **Risk factors:** High doses, rapid dose titration, parenteral administration, multiple concurrent antipsychotic use, dehydration, young age, male sex, and history of NMS
- **Treatment:** Discontinue antipsychotic and initiate supportive care e.g., hydration, benzodiazepines, considered a medical emergency
  - Dantrolene or bromocriptine have been shown to improve symptoms and reduce mortality rates
- Prior to restarting any antipsychotic (ideally a different agent with preference given to SGAs), patients should wait and be monitored for 2 weeks

# SWITCHING BETWEEN ANTIPSYCHOTICS

This tapering regimen is only a guideline → SwitchRx is a great resource  
Use patient-specific factors and adjust algorithm accordingly



## Clinical Tips

- Watch for changes in drug/dosage-related side effects and monitor accordingly
  - For example, antihistaminic effects may cause an increase in appetite
- Taper for 1-2 weeks while starting a new medication

# SPECIAL POPULATIONS - PREGNANCY AND LACTATION

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 The onset of a first episode of psychosis during pregnancy is considered a **psychiatric emergency** and requires treatment

---

- Women with schizophrenia have 2x the rate of adverse outcomes including **stillbirth, infant death, prematurity and small for gestational age**
  - A **consultation with a psychiatrist is recommended** to evaluate the risks/benefits of maintenance antipsychotics during pregnancy
- There have been **no randomized controlled trials of FGAs or SGAs in pregnant patients**
- **Teratogenic effects, including major congenital malformations, have not been demonstrated with the use of antipsychotics during pregnancy**

It is recommended that maintenance antipsychotics during pregnancy are administered at the **lowest dose possible to prevent relapse**

- **Antipsychotics enter breast milk. Olanzapine is preferred choice in lactation, avoid clozapine due to risk of neutropenia and seizures**

# MONITORING PARAMETERS

| Efficacy Endpoints                                            |                  |                                                                                                                                   |
|---------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Parameter                                                     | Degree of Change | Timeframe                                                                                                                         |
| Agitation, hostility, combativeness, anxiety, disrupted sleep | Decrease         | 7 days                                                                                                                            |
| Symptom improvement                                           | Improved         | 4-8 weeks ( <i>change drug if no improvement</i> )<br>6-12 months ( <i>gradual improvement in negative symptoms/functioning</i> ) |

| Safety Endpoints   |                     |                     |
|--------------------|---------------------|---------------------|
| Parameter          | Degree of Change    | Timeframe           |
| NMS                | Prevent             | Duration of therapy |
| Specific Drug ADRs | Prevent or minimize | Duration of therapy |



Note: discontinuation of antipsychotic should be done over **6-24 months**

# SUMMARY OF THERAPY COMPARISON

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1. Clozapine has evidence of superior efficacy for psychosis
2. EPS: highest risk with Risperidone, Paliperidone
3. Anticholinergic effects, cognitive effects and sedation:  
Clozapine > Olanzapine = Quetiapine > Risperidone
4. Weight gain:  
Clozapine > Olanzapine > Quetiapine > Risperidone
5. Sexual Dysfunction: Possible with all antipsychotics

# TAKEAWAY/THERAPEUTIC TIPS

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- **Early detection** of possible psychosis, including recognition of prodromal psychosis and referral to a specialized early psychosis program, is **associated with better symptomatic and functional outcomes**
- **Initiate treatment before the development of a crisis**, such as self-harm, aggression or violence
- **Integrate psychosocial interventions**, such as patient and family psychoeducation, supportive therapy, motivational interviewing, and stress management, with the use of antipsychotic medications to promote adherence to treatment and optimize outcomes
- Conduct **baseline and regular ongoing assessments of signs and symptoms**, possible comorbid conditions (e.g. depression, anxiety, substance use), risk of suicide and aggression, level of functioning, response to treatment, medication side effects, and adherence during all phases of the disorder
  - A variety of standardized scales and semi-structured interviews facilitate these assessments
- To ensure continuity of care, **longitudinal follow-up by the same clinician multidisciplinary team is optimal**



# CASE 1

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LG is a 24-year-old male who was found walking outside in the winter wearing a short-sleeved shirt and shorts. He was brought to the hospital by police after LG's family called and expressed concern for his safety. At the hospital, he appeared to be speaking to himself and had difficulty following directions from the nurses. When asked why he was not dressed appropriately for the weather, he said that he was wearing a special invisible suit that protected him from the cold and from anyone who would harm him. LG's family reported that he had become more and more withdrawn over the past year and that he often appeared suspicious of his family members, accusing them of wanting to kill him. LG does not have a significant past medical history, does not take any medications, and has no allergies. He has a mechanical engineering degree but is currently unemployed.

1. What signs and symptoms support a diagnosis of schizophrenia?
  - a) Withdrawing
  - b) Suspicion of family
  - c) Speaking to self
  - d) All of the above



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2. Which medication can be started for LG?

- a) Haloperidol
- b) Olanzapine
- c) Ziprasidone
- d) Any of the above



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3. If LG has not been sleeping at night, which medication would be best?

- a) Aripiprazole
- b) Loxapine
- c) Lurasidone
- d) Quetiapine



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4. If LG is obese (BMI 31 kg/m<sup>2</sup>), which of the following medications should be avoided the most?

- a) Chlorpromazine
- b) Olanzapine
- c) Paliperidone
- d) Risperidone



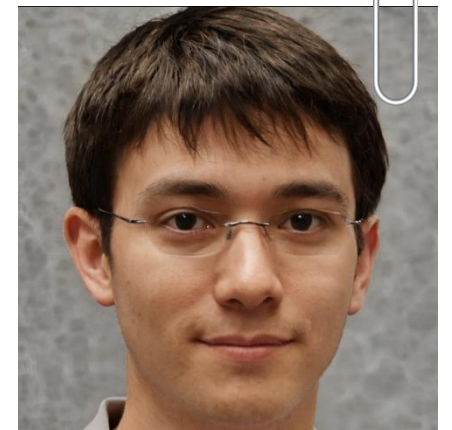
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5. If LG experiences EPS when given a PRN antipsychotic, which medication should be avoided?

- a) Asenapine
- b) Haloperidol
- c) Methotrimeprazine
- d) Quetiapine



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6. If LG has a cardiac abnormality with prolonged QT, which medication should be avoided?

- a) Aripiprazole
- b) Olanzapine
- c) Risperidone
- d) Ziprasidone



# REFERENCES

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# CHANGE LOG

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## August 2025

- Formatting changes made throughout
- Removed table of contents
- Slide 11: Added "Pair with non-pharm measures"
- Slide 12: Added "conventional" to typical antipsychotics and greater serotonin affinity to atypical antipsychotics
- Slide 15: New
- Slide 16 and 17: Added more information to drugs
- Slide 19: New
- Slide 20: Added information to drugs
- Slide 36: References updated

## September 2025

- Slide 33: Rephrased the question and changed option D to align with MCQ exam format

## November 2025

- Slide 12: Added note that all antipsychotics are associated with mortality and stroke in patients with dementia
- Slide 20: Removed point that brexpiprazole use increases mortality risk in the elderly with dementia as this is common for all antipsychotics, not only brexpiprazole
- Slide 21: Removed diltiazem from inhibitors

## March 2026

- Slides 12 and 14: Removed mention of trifluoperazine and thiothixene as not on the Canadian market anymore
- Slide 23: Simplified table
- Slide 28: Added note on sexual dysfunction and updated info on EPS